

Trust and Safety Audit Intention Document

Taaha Bin Mohsin (tb3486)
MS Computer Science
tb3486@nyu.edu

Amory Gao (axg212)
BS/MS Computer Science
axg212@nyu.edu

Mohamed El Atoubi (me2890)
PhD Cybersecurity
me2890@nyu.edu

1 Project Options

- Option One: Original Research Idea
- **Option Two: Platform Measurement Project (Select a platform and harm type, develop an automated detector, and analyze the system.)**

Working Title: Between the Lines: Measuring the Detection Gap for Covert Hate Speech on Reddit

2 Project Summary

Automated content moderation systems have made significant progress in detecting overt hate speech such as direct slurs, explicit threats, and clearly derogatory language. However, bad actors continuously adapt their tactics to evade these systems. One of the most persistent evasion strategies is the use of covert hate speech: coded language (“dogwhistles”), leetspeak substitutions (e.g. the replacement of letters with numbers or symbols), emoji-based encoding, Unicode homographs, and in-group terminology that carries hateful meaning recognizable to a target audience but opaque to automated classifiers. Mendelsohn et al. define dogwhistles as “coded expressions that simultaneously convey one meaning to a broad audience and a second one, often hateful or provocative, to a narrow in-group” [2], deployed specifically to evade both sociocultural repercussions and algorithmic content moderation. In their work, Mendelsohn et al. developed a typology of dogwhistles, curated the largest glossary of over 300 coded terms with rich contextual annotations, and demonstrated that harmful content containing these expressions avoids toxicity detection by standard classifiers. Sasse et al. build on this foundation by proposing methods for discovering emergent dogwhistles through analysis of common online habitats, showing that coded language is not static but evolves dynamically across communities as users adapt to moderation pressure [3]. Overall, prior work suggests there is still a significant gap between evasion tactics

and what automated moderation systems can detect.

This project investigates the following research questions:

1. What proportion of covert hate speech (dogwhistles, leetspeak, and coded language) in polarized Reddit communities is missed by leading off-the-shelf detection tools, specifically Google’s Perspective API and OpenAI’s Moderation API?
2. What are the most common evasion strategies used in Reddit communities to circumvent automated hate speech detection, and which strategies are most effective at evading current tools?
3. Can a custom preprocessing layer, combining a dogwhistle lexicon, text normalization heuristics, and pattern-matching rules, significantly reduce the detection gap when applied before the same off-the-shelf classifiers?

Our motivation stems from the real-world consequences of this detection gap: when covert hate speech evades moderation, it enables sustained harassment of marginalized communities while maintaining deniability. This is distinct from prior work on Reddit abusive language detection, such as general-purpose toxicity classifiers, because we are not building a new hate speech detector. Instead, we are auditing existing detectors to measure where they fail and engineering a targeted pre-processing intervention to attempt to close the gap. We expect to find that off-the-shelf tools miss a substantial proportion of covert hate speech and that certain evasion strategies are more effective than others. Our hope is that even a relatively simple normalization-and-lexicon layer can noticeably improve detection rates without requiring model retraining, though we acknowledge this is an empirical question our project will need to answer.

3 Connection to Trust & Safety Themes

This project connects to multiple core themes from the Trust and Safety syllabus, drawing on both course readings we have already engaged with and upcoming lectures that our project will intersect with as the semester progresses.

The reading that most directly shaped our project idea was Gorwa et al., assigned for the Content Moderation lectures. Their paper lays out how algorithmic moderation systems work, drawing a useful distinction between matching-based approaches (like hash databases) and prediction-based classifiers (like toxicity scoring models). They argue that despite the usefulness of automated techniques, algorithmic moderation threatens to increase opacity, complicate issues of justice, and obscure the political nature of speech decisions. This paper also discusses Google’s Perspective API, which happens to be one of the two tools we are auditing, and notes that observers have constructed adversarial examples that expose its limitations, including the finding that the term “Arabs” was classified as 63% toxic while “I love führer” scored only 3% [1]. Our project provides exactly the kind of empirical audit that Gorwa et al. call for: a systematic measurement of where prediction-based classifiers fail when confronted with adversarial, evasion-oriented content.

The abuse vectors framework from Stephenson et al., which we discussed during the Abuse Vectors lecture (02/09), also shaped how we think about this problem. Their framework is about IoT-enabled abuse, which is obviously a different domain, but the core idea that harmful actors repurpose things that were designed for legitimate use applies surprisingly well to coded hate speech. Dogwhistles are essentially linguistic repurposing: ordinary words get weaponized to carry hateful secondary meanings, and automated systems can’t tell the difference [4]. Just as Stephenson et al. categorize abuse by covertness, ownership, and functionality, our project categorizes evasion strategies by their mechanism (leetspeak, dogwhistles, homoglyphs) and measures how effectively each exploits the gap in current detection systems.

Thomas et al., from the Hate & Harassment reading, provide a taxonomy of online hate and harassment that identifies seven distinct attack categories. Their category of toxic content, which includes hate speech, bullying, and threats, is the attack class our project directly addresses. Importantly, Thomas et al. note that toxic content attackers require no advanced capabilities beyond a communication channel, making it one of the most accessible and scalable forms of online abuse. Their survey data showing that LGBTQ+ individuals face 1.86 times higher odds of experiencing harassment, and that young adults aged 18–24 are 3.99 times more at risk than those 65 and older [5], makes the detection gap we are measuring feel especially urgent. If

the tools meant to protect these communities systematically miss the coded language used against them, that is not just a technical shortcoming. This gap has real consequences for marginalized communities.

Building on these existing readings, our project also connects to upcoming course themes. The two-part exploration of Red Teaming (Part I, 03/23: “Under consensual friendly fire”; Part II, 03/30: “Prompt me if you can”) is relevant and important to our methodology. The first phase of our project is a red-teaming exercise since we are taking existing, deployed content moderation systems and systematically probing them with adversarial content to identify their failure modes. We are not building said systems or training our own classifier, but rather attacking them with real-world evasive content to expose vulnerabilities. Our second phase, building a normalization layer that improves these tools’ performance, shifts from adversarial evaluation to mitigation. Ultimately, we will use this to propose a concrete engineering response to the vulnerabilities we uncover. In a sense, we are trying to follow the full red-teaming cycle to find the weaknesses first, then propose fixes.

Looking ahead, our project will also connect to several upcoming course themes, including Red Teaming (03/23–03/30). Our methodology of systematically probing existing classifiers with adversarial content aligns with the structured adversarial evaluation discussed in these lectures. Our project will also connect to Cross-Cultural and At-Risk Populations (04/06), given that dogwhistles disproportionately target specific marginalized communities, as well as the Special Focus on Hate & Harassment (04/20), which directly addresses the transformation of hateful speech into systemic patterns.

4 Research or Measurement Plan

Guiding Research Questions: Our three research questions (stated above) guide the study: (1) quantifying the detection gap for covert hate speech in leading off-the-shelf classifiers, (2) cataloging and categorizing the evasion strategies that produce this gap, and (3) evaluating whether a custom preprocessing layer can meaningfully close it.

Harm Type: Harassment and hate speech, specifically covert, evasion-oriented hate speech that uses coded language, character substitution, or specialized group-specific terminology to target marginalized groups while evading automated detection.

Platform: Reddit, focusing on English-language communities known for politically polarized or contentious discourse. We will sample from subreddits across a spectrum of moderation intensity and political orientation, selecting

specific communities during the data collection phase based on preliminary exploration. We will also include well-moderated control subreddits (e.g., r/AskHistorians, r/science) where covert hate speech is expected to be rare, to calibrate our detection tools against a stable low-prevalence baseline.

Data Collection: We will collect post and comment data using PRAW (Python Reddit API Wrapper)¹, a mature, open-source Python library providing authenticated access to Reddit’s free Data API. Reddit permits academic, non-commercial use at no cost with a rate limit of 100 requests per minute, which should be more than sufficient for our study². API credentials are obtained instantly by registering an application at reddit.com/prefs/apps. For each target subreddit, we will collect 100–200 recent posts and their top-level comment threads, sorted by “hot,” “top,” and “controversial” to capture a range of engagement levels. Our target is a raw dataset of approximately 2,000–3,000 posts and comments across all subreddits, from which we will draw a stratified sample for annotation. For each item, PRAW provides: title, body text, score, upvote ratio, comment count, awards, timestamp, author (anonymized in our dataset), and subreddit. All data will be collected from publicly posted content; no private messages or user account data will be collected.

Automated Detection Approach. Our methodology centers on a three-way comparison designed to quantify the detection gap and evaluate our proposed intervention:

- **Component 1 – Off-the-Shelf Baselines (Red-Teaming Phase):** We will pass every collected post and comment through two leading, freely available content moderation APIs: Google’s Perspective API³, which returns toxicity, severe toxicity, identity attack, insult, and threat scores on a 0–1 scale; and OpenAI’s Moderation API⁴, which classifies content across categories including hate, hate/threatening, harassment, and self-harm. Both APIs are free, require only an API key to access, and support high request volumes (Perspective API permits about 50 queries per second). For each item, we will record all returned scores and flag items above standard toxicity thresholds (e.g., 0.7 for Perspective API). By comparing these automated classifications against our human-annotated gold standard, we directly measure the detection gap, namely identifying the specific items that contain hate speech but that existing tools fail to flag.

- **Component 2 – Custom Normalization + Lexicon**

¹<https://praw.readthedocs.io>

²<https://support.reddithelp.com/hc/en-us/articles/14945211791892>

³<https://perspectiveapi.com>

⁴<https://platform.openai.com/docs/guides/moderation>

Layer (Mitigation Phase): We will build a preprocessing pipeline that normalizes evasion tactics before passing text to the same off-the-shelf classifiers. This layer has three sub-components: (a) a text normalizer that decodes leetspeak substitutions (e.g., “1” instead of “l”, “3” instead of “e”, “0” instead of “o”), strips inserted punctuation and whitespace (“h.a.t.e” instead of “hate”), and resolves Unicode homoglyphs to their ASCII equivalents, (b) a curated dogwhistle and coded-term lexicon sourced from the glossary of Mendelsohn et al. [2] (300+ terms with contextual annotations), the Anti-Defamation League (ADL) Hate Symbols Database⁵, Hatebase⁶, and supplemented by terms from Sasse et al. [3], mapping coded terms to their underlying hateful meanings, and (c) pattern-matching rules for common structural evasion patterns such as deliberate misspellings, asterisk censoring, and emoji substitutions for slurs. The normalized text will be passed through the same Perspective and OpenAI Moderation APIs, and we will compare detection rates before and after normalization.

- **Component 3 – Human-Annotated Gold Standard:**

Two team members will independently annotate a stratified subset of 100–200 posts and comments. Each item will be labeled along two dimensions: (1) hate speech classification (“Overt Hate”, “Covert Hate”, “Borderline/Ambiguous”, “Not Hateful”) and (2) evasion strategy employed, if any (leetspeak, dogwhistle/coded term, deliberate misspelling, emoji substitution, Unicode homoglyph, punctuation insertion, none). This gold standard will serve as the ground truth for evaluating all automated approaches.

Evaluation. We will compute precision, recall, and F1-score for each detection approach (Perspective API alone, OpenAI Moderation API alone, each API with our normalization layer applied) against the human-annotated gold standard. We will break down results by hate speech type (overt vs. covert) and by evasion strategy, enabling us to identify which specific strategies are most effective at defeating each classifier and which are most improved by our normalization layer.

5 Anticipated Challenges or Risks

Dogwhistle Boundary: The whole point of a dogwhistle is that it can pass as normal speech. Most terms on any dogwhistle glossary are words that are used innocently far more often than they are used hatefully. Our codebook must establish clear decision rules when it comes to posts where dogwhistle signals are unclear. When this happens during the establishment of our human-annotated gold standard,

⁵<https://www.adl.org/hate-symbols>

⁶<https://hatebase.org>

posts will be assigned to a "Borderline/Ambiguous" category. In our detection gap analysis, Borderline/Ambiguous items will be excluded from the primary precision and recall calculations and reported separately, so that our core findings reflect only cases where human annotators reached clear agreement. We will also report the proportion of items classified as Borderline/Ambiguous as a measure of the inherent difficulty of covert hate speech classification, since it is a finding that is itself relevant to the Trust and Safety discussion as it quantifies the gray zone that both human moderators and automated systems have to navigate.

Lexicon Completeness: Conventionalization of existing hate speech, abandonment of algorithmically detected terms, and repurposing of neutral words into dogwhistles are nonstop processes in online hate communities. What we define as a dogwhistle lexicon is a snapshot in time and not a comprehensive glossary. We will document the construction process of our lexicon in full, so that we can properly report our data collection time frame in our final write-up. We expect this to be one of the trickier parts of the project. Our lexicon will inevitably miss terms that emerge after we compile it, and may include terms that have fallen out of use. This is a known limitation, not something we can fully solve.

Researcher Exposure to Hateful Content: We are prepared to engage with large amounts of hate speech, slurs, and dehumanizing rhetoric that could occasionally target communities we identify with. We will mitigate psychological risks through structured annotation shifts, mandatory breaks, and regular team check-ins.

Selection bias: The subreddits we choose to sample from will determine what evasion strategies and language patterns we observe. We will sample from a variety of subreddits, from hate-adjacent communities to intensely moderated and relatively neutral spaces like r/science. We will document our subreddit selection criteria transparently. Our findings can be interpreted as a characterization of detection gaps in the communities we sampled from, and not as a universal account of covert hate speech on Reddit as a whole. If we only look at the worst corners of Reddit, we will find a lot of covert hate speech and overestimate the problem. Conversely, if we only look at well-moderated spaces, we will find almost none. The mix matters, and we will be transparent about it.

6 Expected Outcomes

By the end of the semester, our team expects to deliver the following:

- A human-annotated dataset of 100-200 Reddit posts and comments labeled for overt hate, covert hate, borderline,

and non-hateful content, with evasion strategy annotations

- A taxonomy of evasion strategies observed in Reddit communities, with frequency counts, representative examples (anonymized and paraphrased), and an assessment of each strategy's effectiveness at evading Perspective API and OpenAI's Moderation API.
- A quantified detection gap analysis reporting precision, recall, and F1 for each off-the-shelf classifier on overt versus covert hate speech, showing the magnitude of the gap.
- A custom normalization and lexicon preprocessing pipeline (open-source Python code) with documented improvement in detection rates when paired with the same off-the-shelf classifiers.
- A final report synthesizing findings, situating them within the Trust and Safety literature on adversarial content moderation, and offering evidence-based recommendations for how platform moderation teams could address the covert hate speech gap, including specific technical interventions and policy considerations.
- A presentation summarizing our methodology, key findings, and recommendations

We hope this work helps platforms and researchers better understand where automated hate speech detection falls short, and what relatively low-cost steps could be taken to close the gap.

7 Division of Roles

The current plan is for Taaha to lead pipeline development, including the design and implementation of the normalization layer (leetspeak decoder, homoglyph resolver, punctuation stripper), integration with the Perspective API and OpenAI Moderation API, and construction of the evaluation framework for computing precision, recall, and F1 scores. He will also be responsible for optimizing the preprocessing pipeline based on the false positives and false negatives that emerge. We expect the leetspeak normalization component to be the most straightforward to implement, since it follows mechanical substitution rules. The dogwhistle detection component will likely be harder, since it requires contextual judgment that is difficult to encode in pattern-matching rules.

Amory will lead data collection and statistical analysis, including writing the PRAW-based collection scripts, designing the subreddit sampling and stratification strategy, conducting the statistical analysis of detection gap results, and creating data visualizations and figures for the final report.

Mohamed will lead research design and threat modeling, including research question formulation, dogwhistle lexicon curation from the Mendelsohn et al. glossary, the ADL Hate Symbols Database, Hatebase, and Sasse et al., and the development of the annotation codebook. He will serve as the lead gold-standard annotator, and take primary responsibility for the literature review and final report writing.

All team members will participate in annotation. All members will also contribute to final report editing and presentation preparation. Weekly check-ins will be held to ensure the team is on track, and to address any emerging challenges, including researcher wellbeing concerns arising from sustained exposure to hateful content.

References

- [1] GORWA, R., BINNS, R., AND KATZENBACH, C. Algorithmic content moderation: Technical and political challenges in the automation of platform governance. *Big Data & Society* 7, 1 (2020).
- [2] MENDELSON, J., LE BRAS, R., CHOI, Y., AND SAP, M. From dogwhistles to bullhorns: Unveiling coded rhetoric with language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023).
- [3] SASSE, K., AGUIRRE, C., CACHOLA, I., LEVY, S., AND DREDZE, M. Making FETCH! happen: Finding emergent dog whistles through common habitats.
- [4] STEPHENSON, S., ALMANSOORI, M., EMAMI-NAEINI, P., HUANG, D. Y., AND CHATTERJEE, R. Abuse vectors: A framework for conceptualizing IoT-enabled interpersonal abuse. In *Proceedings of the 32nd USENIX Security Symposium* (2023).
- [5] THOMAS, K., AKHAWA, D., BAILEY, M., BONEH, D., BURSSTEIN, E., CONSOLVO, S., DELL, N., DURUMERIC, Z., KELLEY, P. G., KUMAR, D., MCCOY, D., MEIKLEJOHN, S., RISTENPART, T., AND STRINGHINI, G. SoK: Hate, harassment, and the changing landscape of online abuse. In *2021 IEEE Symposium on Security and Privacy (SP)* (2021).